

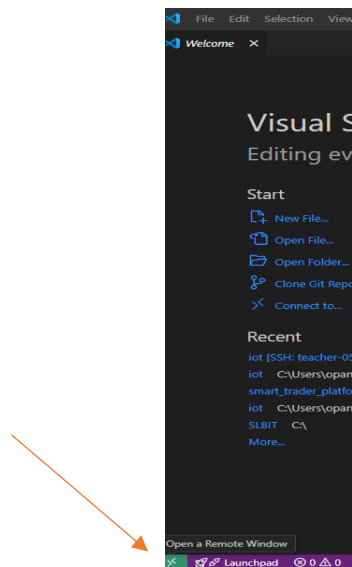
IoT-Lab 02 – Due date: 17 Sep 2026 – 10 points

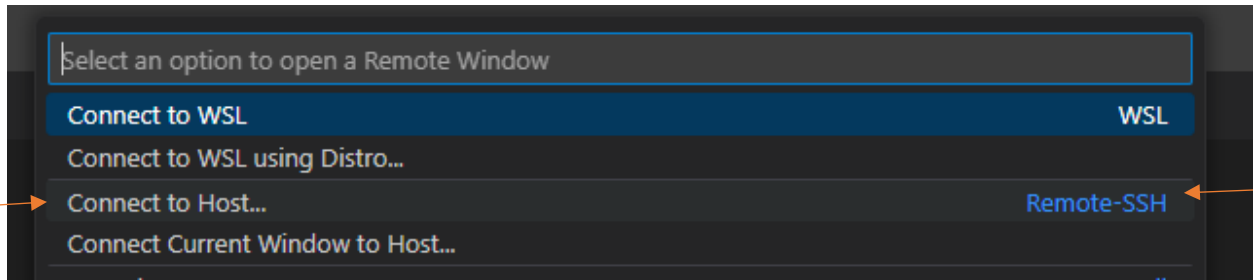
Instructions:

Subject: Publishing IoT data onto the ADAFRUIT IO server

General Instructions:

1. For every Python file that you implement, write your name and id in the first line as comment. Python comments are recommended.
3. You need to show the actual result of this lab to the teacher to be graded.
4. To get the full mark, you must demonstrate your work in the class (even if it is not completed yet) AND then submit the complete version before the deadline. If you missed a class (hence, did not demo your code) and only submitted the code, you will get 50% of the mark. If you did demo your code, but late your will get 50%.
5. You have multiple options to code inside Raspberry pi device. I use visual Studio code **remote server**





6. BE SURE INDISDE YOUR Raspberry you have virtual environment for this LAB something like .venv

```
• (.venv) iot@teacher-05:~/Desktop/iot/lab-02 $ ls -la
total 12
drwxrwxr-x 3 iot iot 4096 Sep 14 06:39 .
drwxrwxr-x 3 iot iot 4096 Sep 14 06:38 ..
drwxrwxr-x 5 iot iot 4096 Sep 14 06:39 .venv
```

7. Use requirements.txt to install dependencies
8. **TASK-01:** complete the Sign up in Adatrut.io . Follow all the the steps. AdaFruit.io works as a monitor server for your simulated sensors
9. **TASK-02:** In this lab, we will run and interact with a Python server program that will let us collect sensor data and view it online from a web browser. We will use the Adafruit IO server for this lab. Adafruit IO is a cloud service that allows you to connect your projects to the internet. It's used for logging and retrieving data, as well as for controlling hardware remotely. You can send data from your sensors to Adafruit IO, visualize it, and then use that data to trigger actions or control other devices connected to the Raspberry PI. It's pretty handy for IoT projects!

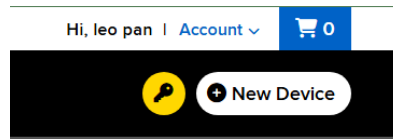
TASK-01

Adafuit server setup:

To get your ADAFRUIT_IO_KEY and ADAFRUIT_IO_USERNAME, just follow these steps:

1. **Go to Adafruit IO:** Open your web browser and navigate to io.adafruit.com.
2. **Sign In or Sign Up:** If you don't have an account, you'll need to sign up. If you already have one, just sign in.

3. **Find Your Key:** Once you're logged in, look for a **yellow key icon** in the top right corner of the page. Click on it.



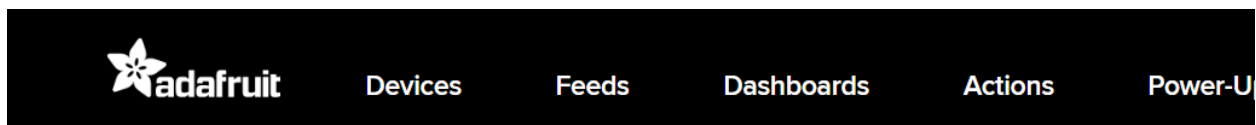
4. **Copy Credentials:** A pop-up window will appear showing your:

- **Username**
- **Active Key** (this is your ADAFRUIT_IO_KEY)

Copy both of these values and paste them into your Python script where it says 'YOUR_AIO_KEY' and 'YOUR_USERNAME'.

Let me know if you have any trouble finding them!

5. **Create feeds for your data points:** click on Feeds menu and then “add feed”



hhichri / Feeds



Default		
Feed Name	Key	Last value
☐ humidity	humidity	42.1
☐ pressure	pressure	1018.05
☐ temperature	temperature	27.8

Add the feeds with names: temperature, pressure, and humidity

Task-02

IoT simulator app

You are given the IoT simulator app (the initial code given) that generates random data for environmental data, security data, and device status data. Modify the given IoT simulator app so that it sends the sensor data to the cloud Adafruit service. Use the knowledge about Adafruit server above.

Here is a initial code which reads data from a file `environmental_data.txt` and send it to AdaFruit IO:

```
## This is the file that simulate 3 sensors
import json
import requests
import time
# complete the code to read config file and return a complete json
def load_credentials(filename="config.json"):
    pass

creds = load_credentials()
ADAFRUIT_IO_USERNAME = creds["adafruit_io_username"]
ADAFRUIT_IO_KEY = creds["adafruit_io_key"]
FEEDS = {
    "temperature": "temperature",
    "humidity": "humidity",
    "pressure": "pressure"
}

...

This function read line by line environment_data.txt

Then store the data inside an array like data=[]

...

def read_json_data(filename):

    pass

def send_to_adafruit_io(feed_name, value):
```

```

url = f"https://io.adafruit.com/api/v2/{ADAFRUIT_IO_USERNAME}/feeds/{feed_name}/data"

headers = {"X-AIO-Key": ADAFRUIT_IO_KEY}

data = {"value": value}

response = requests.post(url, headers=headers, json=data)

return response.status_code == 200
def main():
    # Read your JSONL file
    sensor_data = read_json_data('environmental_data.txt')
    # Send each reading to respective feeds

if __name__ == "__main__":
    main()

```

Example JSON file ('environmental_data.json):

```

[
{"timestamp": "2026-09-14T17:54:53.173622", "temperature": 23.3, "humidity": 55.6, "pressure": 1005.68}
{"timestamp": "2026-09-14T17:54:55.133711", "temperature": 24.2, "humidity": 53.3, "pressure": 1010.66}
{"timestamp": "2026-09-14T17:54:55.134728", "temperature": 24.4, "humidity": 47.5, "pressure": 1014.34}
{"timestamp": "2026-09-14T17:54:55.134728", "temperature": 24.5, "humidity": 46.2, "pressure": 1018.29}
]

```

The output in the console should look as follows:

```

(.venv) iot@teacher-05:~/Desktop/iot/lab-02 $ python sensor_sim.py
Processing reading from 2026-09-14T17:54:53.173622
  Humidity: 55.6% ✓
  Pressure: 1005.68 hPa ✓
Processing reading from 2026-09-14T17:54:55.133711
  Humidity: 53.3% ✓
  Pressure: 1010.66 hPa ✓
Processing reading from 2026-09-14T17:54:55.134728
  Humidity: 47.5% ✓
  Pressure: 1014.34 hPa ✓
Processing reading from 2026-09-14T17:54:55.134728
  Humidity: 46.2% ✓
  Pressure: 1018.29 hPa ✓
Processing reading from 2026-09-14T17:54:55.135725
  Humidity: 50.6% ✓
  Pressure: 1020.55 hPa ✓
Processing reading from 2026-09-14T17:54:56.580494
  Humidity: 48.4% ✓
  Pressure: 1019.24 hPa ✓
Processing reading from 2026-09-14T17:54:56.580494
  Humidity: 49.5% ✓
  Pressure: 1017.45 hPa ✓

```

Figure 1: Output of the simulator code.

After running the code for some time, you should be able to see the data on the Adafruit server:

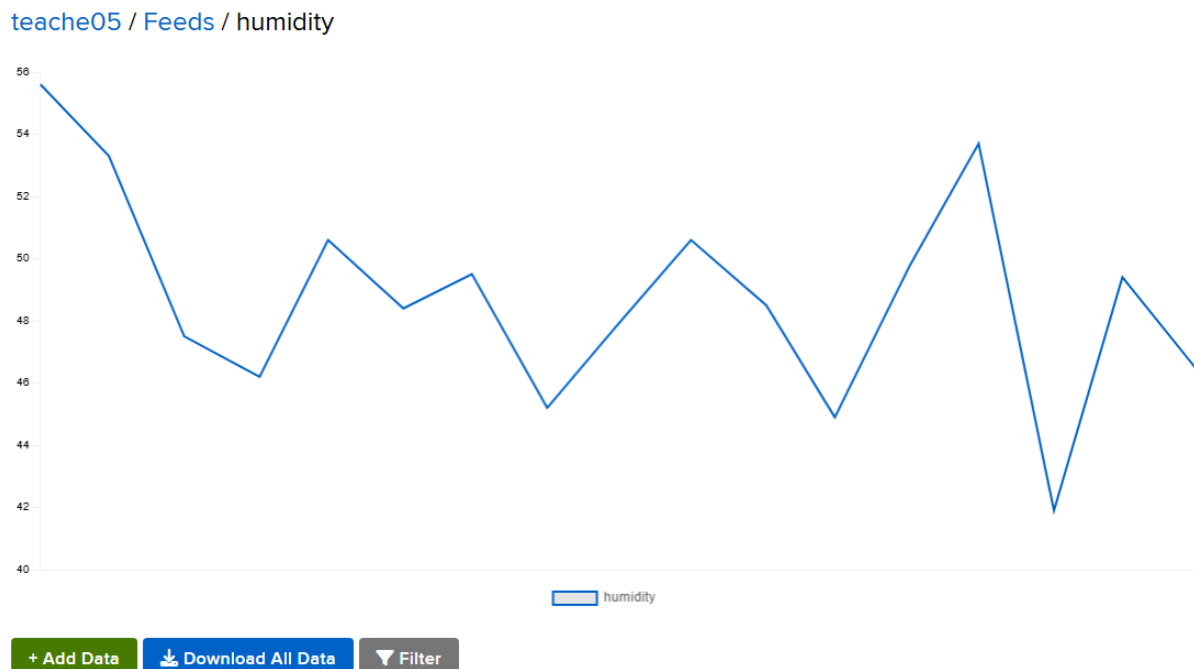


Figure 2: Chart on the ADAFRUIT IO server..